Data Science Assignment

This case consists of a supervised learning example, similar to what we will be working on at Navi.

Your task is to predict the probability of default for the data points where the default variable is not set (type = 'oot'). Please filter the data by type = 'train' and use it to build your machine learning model.

The answer should contain the resulting predictions in a csv file with two columns, decision_id and predicted PD (probability of default==1).

The model should be developed in high-level programming languages like Python, Spark, Java etc. (We don't accept any solutions developed in statistical languages like R and SAS).

Besides the actual results, the following aspects are important:

- **Code quality** (e.g. Use pre-processing pipelines, etc.)
- **Exploratory Data Analysis** – using statistical methods, tell us 2 or 3 things about the data
- **Pre-processing of data and Feature Selection** – You can cover any pre-processing steps required for model building at this stage (if no pre-processing required, do mention your reasoning for the same).
- **Model Building** – You can select one or more model building algorithms for building the model at this stage. Please explain your reasoning for choosing a particular algorithm and evaluation metric for building the model at this stage. Also, any suggestions on how to improve the model you have built here will fetch you brownie points.
- **Model Evaluation** – This stage will include the use of metrics to evaluate the model you have built. Focus here is on your explanation of why the particular metric makes sense for the model algorithm and the problem at hand, not on the result of evaluation.

We expect you to spend not more than 3 hours on this assignment. Focus of the evaluation of assignment is on the approach you take for different steps and not on the actual model results.

All the best.

Dataset

The data is provided in two attached files.

1) assignment_txn_ledger.csv – A comma-separated CSV containing the raw transaction

ledger. Each row is a transaction on a bank account. Schema:

Column	Type	Description
decision_id	string	Unique identifier for the decision
decision_date	date	Date of the decision
bankAccount	string	Bank and account identifier; one decision_id can have multiple accounts
txnDate	date	Date of the transaction
txnTimestamp	integer	Timestamp of the transaction (epoch milliseconds)
debit	float	Amount debited
credit	float	Amount credited
balance	float	Balance after the transaction

2) assignment_flag_base.csv – A comma-separated CSV with one row per decision. Schema:

Column	Type	Description
decision_id	string	Unique identifier
decision_date	date	Decision date
decision_date_full	datetime	Full timestamp of decision
default	float	Target variable: 1. 0 = default, 0.0 = non-default. Empty for type = 'oot'
type	string	'train' = use for modeling; 'oot' = out-of-time, submit predictions for these

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Need to submit the below as a part of the solution:

- One page write-up explaining the solution and thought process.
- Neatly documented code (preferably notebooks)
- Predictions (CSV with two columns: decision_id, predicted PD for all decision_ids where type = 'oot')
- Model pickle
- Code to read and infer using the model file
- Threshold for the +ve class